

CS 40700 Sprint 1 Planning

Product Name: Yarrow

Team Number: 13

Team:

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Sprint Overview

The goal of Sprint 1 is to establish Yarrow's core system architecture and implement the basic functionality needed for its primary user workflows. This includes authentication and user management, document upload and processing, search, and system administration.

By the end of the sprint, users should be able to create an account, log in, upload one or more documents for processing, and view extracted text alongside the original document. Users should also be able to search processed document content. Administrator users should be able to manage non-administrator accounts and view system resource usage.

Scrum Master: Jaime Xu

Meeting Schedule: Monday 11:00 - 11:30 AM

Risk and Challenges:

Yarrow has a relatively complex architecture involving a Next.js frontend, FastAPI backend services, PostgreSQL, document-processing workers, and vLLM-based inference services. A major challenge during Sprint 1 will be integrating and deploying these components while ensuring that they communicate reliably.

Because this is the first implementation sprint, the team may also discover requirements that require changes to the initial architecture. To reduce this risk, the team will keep system components modular and coordinate integration work throughout the sprint.

Current Sprint Detail

Authentication & Security

User Story 1: Create Account

Owner: Anthony McCrovitz | Total Hours: 6 hrs

Description: As a user, I want to create an account so that I can access Yarrow.

Task	Estimated Hours
Create account registration interface	1.5 hrs
Implement backend account creation	1.5 hrs
Implement email verification flow	1.5 hrs
Validate duplicate/invalid account information	0.5 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I enter valid account information, when I submit the registration form, then I receive an email verification code.
- Given I enter the correct verification code, when verification succeeds, then my account is created.
- Given the email is already registered, when I attempt to create an account, then the system rejects the request.
- Given required information is missing or invalid, when I submit the form, then the system displays an appropriate error.

User Story 4: Edit Login Credentials

Owner: Cheng-En Lee | Total Hours: 3 hrs

Description: As a user, I want to edit my login credentials so that I can update my account email or password.

Task	Estimated Hours
Create account settings interface for updating credentials	1 hr
Implement backend endpoint to update email/username and password	1 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I am logged in, when I access account settings, then I can update my email or password independently.
- Given credentials are updated successfully, when I next log in, then the system requires the new credentials.
- Given that I attempt to update my password, when I enter an incorrect current password, then the system rejects the update and displays an error message.

User Story 2: Login

Owner: Anthony McCrovitz | Total Hours: 4 hrs

Description: As a user, I want to log in to my account so that I can access my documents and Yarrow features.

Task	Estimated Hours
Create login interface	1 hr
Implement backend authentication	1 hr
Connect frontend authentication to backend	1 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I have a verified account, when I enter the correct credentials, then I am logged in.
- Given I enter incorrect credentials, when I attempt to log in, then access is denied.
- Given I successfully log in, when authentication completes, then I am redirected to the main application.
- Given I am not authenticated, when I attempt to access a protected page, then I am redirected to login.

User Story 6: Delete Account

Owner: Cheng-En Lee | Total Hours: 3 hrs

Description: As a user, I want to delete my account so that my data is permanently removed.

Task	Estimated Hours
Create account deletion interface with confirmation prompt	0.5 hr
Implement backend account and associated document deletion logic	1.5 hrs
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I am logged in, when I request account deletion, then I am prompted for confirmation.
- Given I confirm deletion, when processing completes, then my account and associated data are permanently deleted.
- Given my account is deleted, when I attempt to log in, then authentication fails.
- Given I cancel the deletion request, when returning to the interface, then my account remains intact.

User Story 19: Secure Logout

Owner: Anthony McCrovitz | Total Hours: 2 hrs

Description: As a user, I want to securely log out of my account.

Task	Estimated Hours
Add logout functionality to the interface	0.5 hr
Clear authentication/session state	0.5 hr

Task	Estimated Hours
Verify protected routes cannot be accessed after logout	0.5 hr
Unit testing	0.5 hr

Acceptance Criteria:

- Given I am logged in, when I select logout, then my authenticated session is ended.
- Given I have logged out, when I attempt to access a protected page, then I am required to log in again.
- Given logout succeeds, when the application updates, then my authenticated user information is no longer available.

User Story 67: Reset Password

Owner: Anthony McCrovitz | Total Hours: 5 hrs

Description: As a user, I want to reset my password if I forget it.

Task	Estimated Hours
Create forgot-password and reset-password interfaces	1 hr
Implement password reset request and email flow	1.5 hrs
Implement password update logic	1 hr
Handle invalid or expired reset requests	0.5 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I have a registered account, when I request a password reset, then reset instructions are sent to my email.
- Given I have a valid reset request, when I submit a valid new password, then my password is updated.
- Given the reset request is invalid or expired, when I attempt to reset my password, then the request is rejected.
- Given my password was successfully reset, when I log in with the new password, then authentication succeeds.

Non-Functional User Story 6: Secure Authentication Credentials

Owner: Anthony McCrovitz | Total Hours: 3 hrs

Description: As a user, I expect my authentication credentials to be securely stored and transmitted.

Task	Estimated Hours
Implement secure password storage	1 hr
Ensure sensitive authentication data is not exposed through APIs or logs	1 hr

Task	Estimated Hours
Security-focused authentication testing	1 hr

Acceptance Criteria:

- Given an account is created, when the password is stored, then the plaintext password is not directly stored.
- Given credentials are submitted, when authentication occurs, then credentials are transmitted through the application's secure communication channel.
- Given an API response or application log is produced, then sensitive authentication credentials are not exposed.

Document Upload & Processing

User Story 3: View Documents

Owner: Aagam Bapna | **Total Hours:** 6 hrs

Description: As a user, I want to view all documents associated with my account.

Task	Estimated Hours
Create API to retrieve user documents	2 hrs
Display documents on frontend	2 hrs
Connect frontend to backend	1 hr
Unit testing	1 hr

Acceptance Criteria:

- Given I am logged in, when I open my documents, then I see my uploaded documents.
- Given I have no documents, when I open my documents, then I see an empty list.
- Given I am logged in, when documents are loaded, then I only see my own documents.

User Story 7: Upload PDF Document

Owner: Anthony McCrovitz | **Total Hours:** 5 hrs

Description: As a user, I want to upload a PDF document so that its contents can be processed.

Task	Estimated Hours
Create PDF upload interface	1 hr
Implement backend upload endpoint	1 hr
Store uploaded PDF and create document processing record	1 hr
Connect upload to document-processing workflow	1 hr

Task	Estimated Hours
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I select a valid PDF, when I upload it, then the system accepts the file.
- Given the upload succeeds, when I view my document library, then the uploaded document appears.
- Given the file is accepted, when processing begins, then the document receives an appropriate processing status.
- Given an upload fails, when the operation completes, then the system displays an appropriate error.

User Story 8: Bulk Document Upload

Owner: Anthony McCrovitz | Total Hours: 4 hrs

Description: As a user, I want to upload multiple documents at once so that I do not need to submit each file individually.

Task	Estimated Hours
Add multi-file selection to upload interface	1 hr
Implement multiple-file upload handling	1 hr
Track the result of each uploaded file independently	1 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I select multiple valid documents, when I upload them, then each file is accepted for processing.
- Given multiple documents are uploaded, when one upload fails, then the remaining valid files are still accepted.
- Given the upload completes, when I open my document library, then each successfully uploaded document appears.

User Story 11: Processing Failure Status

Owner: Anthony McCrovitz | Total Hours: 3 hrs

Description: As a user, I want to know when document processing has failed.

Task	Estimated Hours
Expose failed processing status from backend	1 hr
Persist failed processing state on the document processing record	1 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given document processing fails, when I view the document, then its status is shown as failed.
- Given a document has failed, when I view its status, then the application clearly distinguishes it from processing and completed documents.
- Given processing succeeds, when the status is updated, then the document is not incorrectly marked as failed.

User Story 37: Reject Unsupported File Types

Owner: Anthony McCrovitz | Total Hours: 2 hrs

Description: As a user, I want unsupported file types to be rejected with an explanation.

Task	Estimated Hours
Add file-type validation	0.5 hr
Display unsupported-file error messages	0.5 hr
Backend validation and testing	1 hr

Acceptance Criteria:

- Given I select an unsupported file type, when I attempt to upload it, then the file is rejected.
- Given the file is rejected, when the error is displayed, then I receive a clear explanation.
- Given I select a supported PDF, when I upload it, then it is not rejected by file-type validation.

User Story 38: Document Metadata

Owner: Aagam Bapna | Total Hours: 6 hrs

Description: As a user, I want to view basic information about my documents.

Task	Estimated Hours
Store document metadata	2 hrs
Create API to retrieve metadata	1 hr
Display metadata on frontend	2 hrs
Unit testing	1 hr

Acceptance Criteria:

- Given I uploaded a document, when I view its details, then I see its filename.
- Given metadata exists, when I view the document, then I see its size and upload date.
- Given page count is available, when I view the document, then I see the correct number of pages.

User Story 39: Rename Documents

Owner: Anthony McCrovitz | Total Hours: 2 hrs

Description: As a user, I want to rename uploaded documents.

Task	Estimated Hours
Add rename control to document library	0.5 hr
Implement document rename API/update	0.5 hr
Connect frontend update and add tests	1 hr

Acceptance Criteria:

- Given I own a document, when I rename it, then the new name is saved.
- Given a rename succeeds, when I return to the document library, then the new name is displayed.
- Given I attempt an invalid rename, when I submit it, then the document retains its previous name.

User Story 42: Cancel Queued Processing Job

Owner: Anthony McCrovitz | Total Hours: 4 hrs

Description: As a user, I want to cancel a queued document-processing job.

Task	Estimated Hours
Add cancel action for queued jobs	1 hr
Implement backend job cancellation	1 hr
Update job status after cancellation	1 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given my document-processing job is queued, when I cancel it, then the job is removed from the active queue or marked canceled.
- Given a job has been canceled, when I view its status, then the application shows the canceled state.
- Given a completed job, when I view it, then the cancel action is not incorrectly available.

User Story 5: Delete Documents

Owner: Cheng-En Lee | Total Hours: 3 hrs

Description: As a user, I want to delete documents associated with my account so that unwanted files are removed.

Task	Estimated Hours
Add delete document UI action	0.5 hr
Implement backend document deletion and storage cleanup API	1.5 hrs
Unit and integration testing	1 hr

Acceptance Criteria:

- Given I own a document, when I select delete, then the document and its extracted data are deleted.

- Given a document is deleted, when I refresh my library, then the document no longer appears.
- Given I do not own a document, when I attempt deletion, then access is denied.

User Story 9: Process Uploaded Documents

Owner: Cheng-En Lee | Total Hours: 8 hrs

Description: As a user, I want to process uploaded documents by parsing their text content.

Task	Estimated Hours
Implement document text parsing worker pipeline	2 hrs
Integrate parsing worker with FastAPI backend	2 hrs
Implement db storage system for extracted components from the document	2 hrs
Unit and integration testing for text extraction	2 hrs

Acceptance Criteria:

- Given a document is uploaded, when processing starts, then the text content is extracted successfully.
- Given parsing completes, when I view the document, then extracted text is available for viewing.
- Given unreadable content, when processing completes, then appropriate warnings or failure states are flagged.

User Story 10: Fault-Tolerant Processing

Owner: Cheng-En Lee | Total Hours: 5 hrs

Description: As a user, I want temporary worker failures to not result in loss of uploaded documents or completed data.

Task	Estimated Hours
Implement UI that allows users to resume document processing if the previous parsing job is interrupted	1.5 hrs
Implement backend logic that accepts resume requests from the frontend and resume parsing documents	1.5 hrs
Integration testing under simulated worker failure	2 hrs

Acceptance Criteria:

- If a worker fails mid-job, when the system recovers, then the source document remains preserved.
- Given partial extraction succeeded before failure, when processing resumes, then completed data is retained.
- Given temporary worker loss, when the system recovers, the user can resume processing the partially parsed documents.

User Story 12: View Individual Processed Documents

Owner: Cheng-En Lee | Total Hours: 5 hrs

Description: As a user, I want to view processed documents individually along with their extracted text content.

Task	Estimated Hours
Build single document viewer UI component	1.5 hrs
Connect document detail API endpoint to frontend viewer	1.5 hrs
Unit testing	2 hr

Acceptance Criteria:

- Given a processed document, when I open its view page, then the original pages are rendered.
- Given extracted text exists, when viewing the document, then extracted text is displayed alongside.
- Given extracted text exists, when the user clicks a toggle button, it can toggle between displaying the original document and the extracted document.

User Story 13: View Document Bounding Boxes

Owner: Jaime Xu | Total Hours: 4 hrs

Description: As a user, I want to view bounding boxes on processed documents that link to extracted text content.

Task	Estimated Hours
Implement bounding box overlay component on document viewer	2 hrs
Map bounding box coordinates to extracted text chunks	1 hr
Unit and integration testing	1 hr

Acceptance Criteria:

- Given a processed document, when bounding box overlay is active, then bounding boxes highlight detected text regions.
- Given a bounding box is selected, when clicked, then the corresponding extracted text block is highlighted.
- Given bounding boxes are disabled, when toggled off, then the document displays cleanly without overlays.

User Story 16: Export Extracted Content

Owner: Daniel Benes-Magana | Total Hours: 3 hrs

Description: As a user, I want to export extracted content as Markdown or plain text for external use.

Task	Estimated Hours
Implement Markdown export formatter	1 hr
Add export download button to document viewer UI	1 hr

Task	Estimated Hours
Unit testing	1 hr

Acceptance Criteria:

- Given a processed document, when I click export as Markdown, then a formatted file is downloaded.
- Given extracted text with headers and lists, when exported, then Markdown formatting preserves document structure.
- Given unprocessed documents, when attempting export, then the action is disabled.

User Story 17: Cloud Storage Integration

Owner: Jaime Xu | Total Hours: 6 hrs

Description: As a user, I want to connect cloud storage (Google Drive/Dropbox) to automatically import documents.

Task	Estimated Hours
Implement Google Drive OAuth2 and import connector	2.5 hrs
Create cloud file picker interface	2 hrs
Unit and integration testing	1.5 hrs

Acceptance Criteria:

- Given I connect my Google Drive account, when I select a file, then it is imported into Yarrow.
- Given an imported document, when import completes, then document processing begins automatically.
- Given revoked cloud access, when attempting import, then the system prompts for re-authentication.

User Story 20: Preserve Original Page Numbers

Owner: Jaime Xu | Total Hours: 3 hrs

Description: As a user, I want original page numbers preserved with each piece of extracted information.

Task	Estimated Hours
Store page number metadata during parsing worker execution	1.5 hrs
Display page numbers in extracted text UI	0.5 hr
Unit testing	1 hr

Acceptance Criteria:

- Given extracted text, when viewed, then each text section displays its source page number.
- Given a multi-page document, when parsed, then page numbers match original document paging exactly.
- Given exported data, when exported to JSON, then page numbers are included in payload attributes.

User Story 21: Structured JSON Output

Owner: Daniel Benes-Magana | Total Hours: 5 hrs

Description: As a user, I want extracted document information represented in structured JSON format.

Task	Estimated Hours
Design and implement JSON schema for extracted document data	2 hrs
Implement API endpoint to download structured JSON	1.5 hrs
Unit and schema validation testing	1.5 hrs

Acceptance Criteria:

- Given a processed document, when JSON format is requested, then valid structured JSON is returned.
- Given structured JSON, when validated, then it includes text content, page numbers, and bounding coordinates.
- Given schema compliance check, when tested, then all JSON payloads adhere strictly to the schema definition.

User Story 18: Share Processed Documents for Collaborative Review

Owner: Jaime Xu | Total Hours: 8 hrs

Description: As a user, I want to share processed documents and their bounding-box data with other users so that we can collaboratively review extracted information.

Task	Estimated Hours
Create document sharing and permissions interface	2 hrs
Implement backend sharing endpoints and access policy logic	3 hrs
Support bounding-box and extracted data viewing for shared users	1.5 hrs
Unit and integration testing for sharing permissions	1.5 hrs

Acceptance Criteria:

- Given I own a processed document, when I share it with another registered user, then that user receives access to view the document.
- Given a shared document is opened by a collaborating user, when viewed, then the original pages, bounding boxes, and extracted data are rendered correctly.
- Given a user without sharing permissions, when attempting to view a document, then access is denied.

User Story 22: Identify Document Regions

Owner: Jaime Xu | Total Hours: 6 hrs

Description: As a user, I want the system to identify document regions such as headers, paragraphs, figures, and tables.

Task	Estimated Hours
Implement layout analysis pipeline for region classification	2.5 hrs
Integrate region labels into document viewer UI	2 hrs
Unit and integration testing	1.5 hrs

Acceptance Criteria:

- Given a processed document, when viewed, then headers, paragraphs, tables, and figures are tagged correctly.
- Given region filtering in UI, when a region type is selected, then corresponding sections are highlighted.
- Given exported JSON, when inspected, then region types are recorded for each content block.

User Story 23: Structure-Preserving Table Extraction

Owner: Daniel Benes-Magana | Total Hours: 6 hrs

Description: As a user, I want tables extracted while preserving their rows, columns, and layout structure.

Task	Estimated Hours
Implement table structure recognition module in processing pipeline	3 hrs
Render extracted tables cleanly in document viewer	1.5 hrs
Unit and extraction accuracy testing	1.5 hrs

Acceptance Criteria:

- Given a document containing tables, when processed, then row and column structure is preserved.
- Given an extracted table, when viewed, then cell text aligns with the original tabular layout.
- Given table data, when exported to Markdown or JSON, then tabular syntax is properly structured.

User Story 24: Reconstruct Multi-Page Tables

Owner: Cheng-En Lee | Total Hours: 6 hrs

Description: As a user, I want tables that span page breaks reconstructed as a single logical table.

Task	Estimated Hours
Implement multi-page table stitching logic	3 hrs
Update viewer to display stitched logical tables	1.5 hrs
Unit testing on multi-page sample documents	1.5 hrs

Acceptance Criteria:

- Given a table continuing across page breaks, when processed, then it is joined into one logical table.
- Given a stitched table, when exported, then rows across pages are preserved sequentially.

- Given original page mapping, when viewing cells, then source page numbers for each row remain traceable.

User Story 25: Side-by-Side Document View

Owner: Cheng-En Lee | Total Hours: 5 hrs

Description: As a user, I want to view original documents and extracted structured data side-by-side.

Task	Estimated Hours
Build split-screen layout component for PDF and extracted data	2.5 hrs
Implement synchronized scrolling across split panes	1.5 hrs
UI unit testing	1 hr

Acceptance Criteria:

- Given a processed document, when opened, then original PDF and structured data display side-by-side.
- Given split-screen mode, when I resize pane divider, then both views scale appropriately.
- Given mobile screen size, when viewing, then layout falls back to single-pane tabbed view.

User Story 26: Cross-Highlighting from Extracted Data

Owner: Jaime Xu | Total Hours: 5 hrs

Description: As a user, I want clicking extracted data to highlight the corresponding region on the original document.

Task	Estimated Hours
Implement click event handlers linking extracted items to bounding box coordinates	2 hrs
Implement PDF canvas auto-scroll to highlighted bounding box	2 hrs
Integration testing	1 hr

Acceptance Criteria:

- Given a side-by-side view, when I click an extracted paragraph, then the PDF scrolls to highlight its source region.
- Given highlighting is active, when clicking another item, then previous highlight clears and new region highlights.
- Given an item on a different page, when clicked, then the viewer automatically turns to that page.

User Story 40: Sort Document Library

Owner: Jaime Xu | Total Hours: 3 hrs

Description: As a user, I want to sort my documents by name, upload date, or processing status.

Task	Estimated Hours
Add sorting control dropdown to document list UI	1 hr
Implement sorting query parameters in backend list API	1 hr
Unit testing	1 hr

Acceptance Criteria:

- Given the document library, when I select sort by name, then documents display in alphabetical order.
- Given the document library, when I select sort by upload date, then the newest documents appear first.
- Given the document library, when I select sort by status, then documents are grouped by processing state.

User Story 41: Filter Library by Processing Status

Owner: Daniel Benes-Magana | Total Hours: 3 hrs

Description: As a user, I want to filter my document library by processing status.

Task	Estimated Hours
Add status filter controls to document library UI	1 hr
Implement backend status filtering API logic	1 hr
Unit testing	1 hr

Acceptance Criteria:

- Given a document library, when I filter by completed status, then only fully processed documents show.
- Given a document library, when I filter by failed status, then only failed processing documents show.
- Given filter cleared, when reset, then all documents display again.

User Story 43: Reprocess Document

Owner: Daniel Benes-Magana | Total Hours: 3 hrs

Description: As a user, I want to reprocess a document without needing to upload it again.

Task	Estimated Hours
Add reprocess button to document action menu	0.5 hr
Implement backend re-queuing trigger endpoint	1.5 hrs
Unit and integration testing	1 hr

Acceptance Criteria:

- Given an uploaded document, when I click reprocess, then a new processing job is created using the existing file.
- Given reprocess starts, when looking at status, then state updates to queued/processing.

- Given reprocessing completes, when viewing details, then updated extraction results replace old data.

User Story 44: Preserve Multi-Column Reading Order

Owner: Daniel Benes-Magana | Total Hours: 6 hrs

Description: As a user, I want the system to preserve correct reading order in multi-column documents.

Task	Estimated Hours
Implement reading order sorting algorithm in OCR pipeline	3 hrs
Validation testing on multi-column document test suite	3 hrs

Acceptance Criteria:

- Given a multi-column document, when processed, then text is extracted column by column sequentially.
- Given extracted text, when read, then sentences across column boundaries do not mix incorrectly.
- Given a single column document, when processed, then standard top-to-bottom reading order is preserved.

Search

User Story 14: Search Documents

Owner: Agam Bapna | Total Hours: 7 hrs

Description: As a user, I want to search my processed documents by text.

Task	Estimated Hours
Create document search API	3 hrs
Connect search to frontend	2 hrs
Unit and integration testing	2 hrs

Acceptance Criteria:

- Given my documents contain a word, when I search for it, then I see matching results.
- Given there are no matches, when I search, then I see no results.
- Given another user's document contains the word, when I search, then their document is not shown.

User Story 27: Search Result Details

Owner: Agam Bapna | Total Hours: 8 hrs

Description: As a user, I want search results to show the document, page, and matching text so I can find the information I need.

Task	Estimated Hours
Add document and page information to search results	2 hrs

Task	Estimated Hours
Add matching text around each search result	2 hrs
Connect search results to the document viewer	2 hrs
Unit and integration testing	2 hrs

Acceptance Criteria:

- Given that matching documents exist, when the user performs a document search, the user should see a list of document names that matches the search.
- Given matching documents exist with multiple hits in a single document, when a user searches, all matches within each document are displayed.
- Given matching documents exist, when a user searches, the page numbers containing matches are displayed.

User Story 15: Advanced Document Search

Owner: Daniel Benes-Magana | Total Hours: 4 hrs

Description: As a user, I want to perform advanced searches across documents using date range and document type filters.

Task	Estimated Hours
Add search filter controls (date range, type) to search UI	1.5 hrs
Implement filtered search backend query handling	1.5 hrs
Unit and integration testing	1 hr

Acceptance Criteria:

- Given a search term and date filter, when executed, then only matching documents within date range return.
- Given a document type filter, when selected, then results restrict strictly to that document type.
- Given no matches satisfy all selected filters, when executed, then the empty results page displays.

User Story 28: Navigate from Search to Source Document

Owner: Daniel Benes-Magana | Total Hours: 4 hrs

Description: As a user, I want to click a search result and navigate directly to the corresponding location in the source document.

Task	Estimated Hours
Connect search result click handler to document viewer routing	1.5 hrs
Implement auto-navigation and bounding box highlight on page load	1.5 hrs
Integration testing	1 hr

Acceptance Criteria:

- Given a search results list, when I click a result, then the viewer opens to that specific document and page.
- Given viewer opens from search result, when loaded, then the search term target region is highlighted.
- Given the back button clicked, when returning, then search context and results remain preserved.

Administration

User Story 70: View User Accounts

Owner: Aagam Bapna | **Total Hours:** 6 hrs

Description: As an admin, I want to view all user accounts.

Task	Estimated Hours
Create admin account API	2 hrs
Display accounts on admin page	2 hrs
Add admin access control	1 hr
Unit testing	1 hr

Acceptance Criteria:

- Given I am an admin, when I open the account page, then I see all accounts.
- Given an account exists, when I view it, then I see its account information.
- Given I am not an admin, when I try to access the account page, then access is denied.

User Story 73: Admin Dashboard

Owner: Aagam Bapna | **Total Hours:** 7 hrs

Description: As an admin, I want to see basic system statistics.

Task	Estimated Hours
Create API for system statistics	2 hrs
Calculate user, document, storage, and job statistics	2 hrs
Display statistics on admin dashboard	2 hrs
Unit testing	1 hr

Acceptance Criteria:

- Given I am an admin, when I open the dashboard, then I see total users and documents.
- Given system data exists, when I open the dashboard, then I see storage and job statistics.
- Given I am not an admin, when I try to open the dashboard, then access is denied.

Backlog

Functional Requirements

Task #	Functional Requirement	Estimated Hours on Task
1	As a user, I would like to create an account	2
2	As a user, I would like to login to the account I have created	1
3	As a user, I would like to view all the documents associated with my account	3
4	As a user, I would like to edit my login credentials	1
5	As a user, I would like to delete documents associated with my account	1
6	As a user, I would like to delete my account	1
7	As a user, I would like to upload a PDF document so that its contents can be processed.	2
8	As a user, I would like to upload multiple documents at once so that I do not need to submit each file individually.	2
9	As a user, I would like to process the documents I have uploaded by parsing their text content.	4
10	As a user, I want temporary worker or processing failures to not result in loss of my uploaded document or previously completed processing data	5
11	As a user, I would like to know when document processing has failed.	2
12	As a user, I want to view processed documents individually and their extracted text content	3
13	As a user, I want to view the bounding boxes on each processed document that links to certain extracted text content.	2
14	As a user, I would like to look up specific pages of processed documents by text search.	4
15	As a user, I would like to perform advanced searches across documents using filters like date ranges or document type.	3
16	As a user, I would like to export extracted content for external use, such as markdown	3

17	As a user, I would like to connect my cloud storage (e.g., Google Drive/Dropbox) to automatically import documents.	5
18	As a user, I would like to share processed documents and their bounding-box data with other users for collaborative review.	8
19	As a user, I would like to securely log out of my account.	2
20	As a user, I would like the system to preserve the original page number associated with each piece of extracted information	4
21	As a user, I would like extracted document information to be represented in structured JSON so that it can be used programmatically	6
22	As a user, I would like the system to identify different document regions such as headers, paragraphs, figures, and tables.	8
23	As a user, I would like tables to be extracted while preserving their rows, columns, and structure.	8
24	As a user, I would like tables that continue across page breaks to be reconstructed as one logical table.	10
25	As a user, I would like to view the original document and its extracted structured data side-by-side.	6
26	As a user, I would like to click extracted data and have the corresponding location highlighted on the original document.	6
27	As a user, I would like search results to show the document, page, and matching extracted content where the result originated.	5
28	As a user, I would like to click a search result and navigate directly to the corresponding location in the source document.	5
29	As a user, I would like failed document-processing jobs to be retried without uploading the document again.	5
30	As a user, I would like to view the progress of a document-processing job while it is running.	4
31	As a user, I would like to view the overall progress of a batch containing multiple documents.	5

32	As a user, I would like to receive warnings that my document may be blurred instead of silently receiving poor extraction results.	3
33	As a user, I would like to receive warnings that my document may be skewed instead of silently receiving poor extraction results.	3
34	As a user, I would like to receive warnings that my document may be cropped/truncated instead of silently receiving poor extraction results.	3
35	As a user, I would like to receive warnings that my document may be too dark/too light instead of silently receiving poor extraction results.	3
36	As a user, I would like to upload PNG and JPEG document scans in addition to PDFs.	5
37	As a user, I would like unsupported file types to be rejected with an explanation.	3
38	As a user, I would like to view document metadata including file name, page count, upload date, and processing status.	4
39	As a user, I would like to rename uploaded documents.	3
40	As a user, I would like to sort my documents by name, upload date, or status.	3
41	As a user, I would like to filter my document library by processing status.	3
42	As a user, I would like to cancel a queued document-processing job.	5
43	As a user, I would like to reprocess a document without uploading it again.	4
44	As a user, I would like the system to preserve the correct reading order in multi-column documents.	10
45	As a user, I would like rotated pages to be detected and corrected before OCR.	8
46	As a user, I would like extracted text to include a confidence score.	6
47	As a user, I would like low-confidence extracted regions to be visually marked for review.	5
48	As a user, I would like figure captions to remain	8

	associated with their figures.	
49	As a user, I would like extracted table data to be exportable as CSV.	5
50	As a user, I would like extracted data to be exportable as JSON.	4
51	As a user, I would like to manually correct incorrectly extracted text.	8
52	As a user, I would like my corrections to be saved.	5
53	As a user, I would like to toggle bounding boxes on and off.	2
54	As a user, I would like bounding boxes to remain aligned while zooming the document.	6
55	As a user, I would like search matches highlighted after navigating to their source page.	4
56	As a user, I would like search results to include a short surrounding text snippet.	4
57	As a user, I would like to perform exact-phrase searches.	4
58	As a user, I would like to search for extracted table contents.	6
59	As a user, I would like to sort search results.	3
60	As a user, I would like to receive a notification when a long-running batch completes.	4
61	As a developer, I would like automated unit tests for critical backend API functionality.	10
62	As a developer, I would like automated tests for OCR and document-processing functionality.	10
63	As a developer, I would like integration tests covering document upload, processing, storage, and retrieval.	12
64	As a developer, I would like integration tests for table and layout extraction using known test documents.	12
65	As a developer, I would like end-to-end tests covering major user workflows.	12
66	As a developer, I would like tests to run automatically when code changes are merged.	12

67	As a user, I would like to reset my password if I forget it.	5
68	As a user, I would like to search my document library by document name	4
69	As a user, I would like to view information about why a document-processing job failed so that I can determine how to correct the problem.	5
70	As an administrator, I would like to view a list of all accounts with name, email, join date, and document count	2
71	As an administrator, I would like to disable a user's account so they cannot log in, while keeping their documents intact.	2
72	As an administrator, I would like to promote a user to administrator	1
73	As an administrator, I would like a dashboard showing total users, documents, storage used, and jobs by state.	3
74	As an administrator, I would like to view processing jobs across all users, including queue position, current stage, and failure reason	4
75	As an administrator, I would like to force-cancel a stuck job and retry a failed job system-wide	3
76	As an administrator, I would like to see the status of each worker node and inference server	3
77	As a user, I would like PDFs that already contain a text layer to skip OCR and reuse the embedded text and positions.	6
78	As a user, I would like to be prompted for a password when uploading an encrypted PDF instead of the job failing silently.	3
79	As a user, I would like to be warned when I upload a document identical to one I already have.	4
80	As a user, I would like to see upload progress per file and retry a failed large upload without starting over.	5
81	As a user, I would like to upload documents by dragging and dropping files or a folder.	3
82	As a user, I would like to see how much of my	3

	storage quota I have used.	
83	As a user, I would like to see processing status for each page, not only the document as a whole.	5
84	As a user, I would like only failed pages to be retried instead of the whole document.	6
85	As a user, I would like each result to record which model version produced it so I can reprocess when models improve.	4
86	As a user, I would like to be warned when a page contains handwriting so I know extraction quality may be lower.	5
87	As a user, I would like the system to detect document language and use the appropriate OCR model.	6
88	As a user, I would like repeated headers, footers, and page numbers to be detected and excluded from paragraphs and tables.	8
89	As a user, I would like footnotes to be linked to the text they reference.	6
90	As a user, I would like figures to be extracted as image files and included in exports.	5
91	As a user, I would like each table cell to have its own bounding box.	6
92	As a user, I would like small jobs to be prioritized so they are not stuck behind a large batch.	5
93	As a user, I would like search to tolerate common OCR errors (such as 1 vs l, 0 vs O).	6
94	As a user, I would like to search within a single document.	3
95	As a user, I would like to filter search results by confidence score.	3
96	As a user, I would like my manual corrections to be reflected in search results.	5
97	As a user, I would like to search table contents by column header.	6
98	As a user, I would like page thumbnails for navigating long documents.	4
99	As a user, I would like keyboard shortcuts in the	3

	viewer (next page, next low-confidence region, toggle boxes).	
100	As a user, I would like to see the difference between the original extraction and my correction, and undo a correction.	6

Non-Functional Requirements

Task #	Non-Functional Requirement	Estimated Hours on Task
1	As a user, I would like to set up two-factor authentication for my account to increase security.	3
2	As an administrator, I would like to scale the document processing service by running more worker nodes and inference servers	5
3	As a user, I want the application to have an intuitive web interface supporting desktop, laptop, and mobile screens.	5
4	As a user, I want the application to be responsive. This means the application should provide appropriate loading indicators or allow me to continue using other parts of the application without interruption.	5
5	As a user, I expect uploaded documents and their extracted contents to be only accessible by authorized users	3
6	As a user, I expect my authentication credentials to be securely stored and transmitted	4
7	As a user, I want search results across processed documents to be returned within a reasonable response time under normal system load	4
8	As a user, I want long-running OCR and document-processing jobs to run in the background so that I can	5

	continue using other parts of the application	
9	As a user, I want the failure of one document processing job to not cause other documents in the batch to fail	4
10	As a user, I want failed processing jobs to be recoverable or retryable without needing to upload the document again	4
11	As an administrator, I want additional worker nodes to be added without redesigning the document processing system so that processing capacity can scale	5
12	As a user, I want clear error messages that explain when an upload, processing job, or search operation fails	4
13	As a user, I want document processing states such as queued processing, completed, and failed to be clearly distinguishable	4
14	As a developer, I want the document processing, search, storage, and interface components to be sufficiently modular so that they can be tested and modified independently	5
15	As a user, I want the application to support current versions of major desktop browsers so that I can access the system reliably across common environments.	4
16	As a user, I want core application workflows to remain usable through keyboard navigation and clearly labeled interface elements.	5
17	As a user, I expect deleting a document to also remove its associated extracted data and search-indexed content.	5
18	As a developer, I want processing failures to generate logs that identify	6

	the affected document, job, and processing stage so that problems can be diagnosed efficiently.	
19	As a developer, I want environment-specific configuration such as credentials, storage locations, and inference endpoints to be configurable rather than hard-coded.	5
20	As a user, I want previously completed document-processing results to remain available even if a worker or inference service temporarily fails.	5
21	As an administrator, I would like to set global limits such as maximum file size and maximum documents per user, so one account can't consume all storage.	3
22	As an administrator, I would like a log of administrative actions, recording who disabled or deleted what and when	3